ResonatE: One Knowledge-Graph Model on Two OGB Leaderboards

Free block operators, a row-sparse training shell, label-free retrieval from the training graph, and distillation at temperature 2, on `ogbl-biokg` and `ogbl-wikikg2`

Cristian Malaia
`cristian.malaia@gmail.com`

Claude Fable 5
Anthropic

September 2026

Preprint, draft of 4 September 2026. Not peer reviewed. Code, logs, receipts and checkpoints: https://github.com/jinxmcg/resonate; project page: https://resonate.page. Five leaderboard entries are being filed from the receipts in the repository.

Abstract

We report one knowledge-graph embedding model, one training shell and one recipe on two Open Graph Benchmark link-prediction tasks of very different shape, with two capacity dials set on validation per graph. Entities are unit-norm vectors of $M = k^2$ complex coefficients stored through their real view with row-sparse gradients and row-wise Adagrad; each relation and direction is a free block-diagonal operator, unitary at initialisation; a hop applies the operator and renormalises; the score is an inner product against the raw target row. Every number below is the mean and unbiased standard deviation over ten seeds with one test read per seed under the official Evaluator. On `ogbl-biokg` (93k typed entities, 51 relations) a 27.1M-parameter model ($k = 12$, 4×4 blocks) reaches test MRR 0.8158 ± 0.0006; with four label-free retrieval features read from the training graph and blended per relation on validation, 0.8463 ± 0.0004; distilled at training time from the ten seeds at temperature 2, 0.8528 ± 0.0002, fifth of twelve on the board and the fourth single model, at the smallest parameter count on it by $3.5\times$. On `ogbl-wikikg2` (2.5M entities, 535 relations) a 329M-parameter model ($k = 8$, one dense 64×64 block per relation) trains in 36 minutes on one RTX 5090 and reaches 0.6676 ± 0.0010; with neighbourhood and analogy members from the training graph, self-proposed edges, and a learned per-relation combiner, the distilled single model reaches 0.7320 ± 0.0010, which would be second of twenty-eight and the best entry without entity text; a ten-seed ensemble under the same combiner reaches 0.7426 ± 0.0002 (leave-one-out), above the current first entry. The sparse shell was built for the large graph, brought back to the small one under a pre-registered bar, and won on every seed (+0.0040); the temperature was found on the large graph and won on the small one the same way (+0.0019). The same table serves training, scoring, the retrieval members, a compiled index whose multi-hop query cost is flat in depth, and the model's own edge proposals. We report every development test read, the two runs that were read twice, and the temperature and block-size curves that fix the dials.

1 Introduction

Knowledge-graph embedding models score a triple (h, r, t) by a function of learned vectors. On the two OGB link-prediction tasks studied here [11], the boards are shaped by different things: `ogbl-biokg` is a dense, typed biomedical graph with 51 relations where the classic bilinear and rotational families (DistMult, ComplEx, RotatE) sit near 0.80–0.81 at roughly 188M parameters and the leaders reach 0.85–0.86 through learned scoring functions, auxiliary objectives or relation-aware ensembles [20]; `ogbl-wikikg2` is a sparse 2.5M-entity Wikidata subset with 535 relations and uniform negatives, where the leaders are node-association encoders and ensembles at 0.72–0.74 [1], and a single 250-dimensional ComplEx scores 0.40.

This report describes one model on both. ResonatE was designed for multi-hop reasoning: an entity is a unit-norm vector, a relation is a linear operator, a hop is "apply the operator and renormalise", and a chain of hops is a chain of operators. The benchmarks are single-edge tasks, so only the one-hop score matters for the boards, but the design leaves marks on the results: renormalisation after every hop makes the usual N3 regulariser redundant, exact operator composition makes path training redundant, type-grouped batches make a relation-prediction loss redundant—each verified negatively on `ogbl-biokg` (Section 6.5)—and the compiled operator chain gives multi-hop queries a latency that is flat in depth (Section 7).

The path from one board to the other is the substance of this report. The first `ogbl-biokg` ladder was trained with dense Adam on the entity table and published as a preprint on 3 September 2026 (0.8118/0.8425/0.8505). Scaling that shell to 2.5M entities is where it stops: Adam streams the whole table and two moments every step. The shell was rebuilt for `ogbl-wikikg2` around a real-view table with row-sparse gradients and row-wise Adagrad [5, 16] ($21\times$ faster per step, identical quality), then brought back to `ogbl-biokg` under a pre-registered bar, where it beat the dense ladder on every seed. The same happened to the distillation temperature: $T = 1$ transferred nothing on the large graph, $T = 2$ transferred everything, and $T = 2$ then improved the small graph too. The rows submitted to both boards are the ones that came out of this loop; the first ladder is kept as history.

Our contributions are the five ten-run results, the two capacity dials and the measured reason they differ between the graphs, the retrieval members and their two blends, the temperature curve, and a fully scripted reproduction path with receipts and released checkpoints for both boards. The operator family itself is not a contribution; the next paragraph places it.

Relation to prior operator families. Representing a relation as a rotation or a small orthogonal block, so that mixed-relation chains compose non-commutatively, is an established line: RotatE [22] uses 1×1 complex rotations (commuting), QuatE [29] and DensE [18] quaternion rotations, Rotate3D [6] rotations in $\mathbb{R}^3$, OrthogonalE [33] block-diagonal orthogonal matrices with Riemannian optimisation and matrix-valued entities, and Knowledgebra [25] free matrix semigroups. ResonatE's operator is nearest to OrthogonalE: block-diagonal with small complex blocks, unitary at initialisation—but the constraint is dropped afterwards (the blocks are free), the query side is renormalised after every hop while the target side keeps its norm as a popularity channel, and no path or composition loss is used. We make no claim that this operator is new or better than those families; on `ogbl-biokg` the block size moves validation MRR by $+0.0055$ ($2 \times 2 \to 4 \times 4$ at 50k steps) and is flat from 4×4 to 16×16, which is the size of the effect the operator form has there. The project's name and the "mode" vocabulary in the code are a coordinate label: the score is invariant under any block-diagonal unitary change of basis ($E \to UE$, $H_r \to UH_rU^\dagger$), no Fourier transform is taken anywhere, and nothing in the results depends on the coordinates being spectral. Nor is "complex" a claim: a complex 4×4 block on $\mathbb{C}^{144}$ and a free real 4×4 block on $\mathbb{R}^{288}$ have the same 27,124,129 parameters, and on the 12.5k-step validation screen with two seeds each the real model scores 0.7923 against the complex model's 0.7897 (a real 8×8 block, which contains the complex 4×4, 0.7898)—below the 0.005 we had fixed as the smallest meaningful gap, with the sign against the complex form. The complex arithmetic is a way of writing a real block model on $\mathbb{R}^{2M}$, and the released code accepts `real=True` to write it the other way.

2 Model

Entities and relations. Let N be the number of entities and $M = k^2$. The entity table is $E \in \mathbb{C}^{N\times M}$. A query entity is read through $\mathrm{cnorm}(E_h) = E_h/\|E_h\|$; a target entity is read as

Table 1: The one recipe and its two settings. Everything not listed is shared: unit-norm query, raw target row, free blocks unitary at initialisation, sampled-softmax loss with a $0.1\|z - E_t\|^2$ trajectory term, gradient clipping at 1, cosine schedules, the training shell of Section 2, the members and blends of Section 3, distillation at $T = 2$ (Section 4).

	`ogbl-biokg`	`ogbl-wikikg2`
entities / relations / training triples	93,773 / 51 / 4.76M	2,500,604 / 535 / 16.1M
negatives (Evaluator)	500, type-matched	500, uniform
k ($M = k^2$)	12 (144)	8 (64)
block size b	4×4 (36 blocks)	64×64 (1 block)
parameters	27.1M	328.8M
batch / sampled negatives	2,048 / 4,096 (type range)	2,048 / 4,096 (uniform)
head-query fraction	0.5	0.75
table learning rate (RowAdagrad)	0.3	0.6
steps, single model	50,000	800,000
steps, student	50,000	400,000
time per seed on one RTX 5090	4 min (student 8.5)	36 min (student 50)

the raw row E_t, so target norms are free and act as a learned per-entity popularity channel. For each of the $2R$ (relation, direction) pairs (both directions are separate operators) the operator H_r is block-diagonal with M/b independent $b \times b$ complex blocks, initialised as random unitaries by QR decomposition and then trained freely. A hop is

$$z = \mathrm{cnorm}(H_r \ \mathrm{cnorm}(E_h)), \tag{1}$$

and the score of candidate t is

$$s(h, r, t) = \mathrm{Re}\langle z, E_t\rangle \cdot e^{\tau}, \tag{2}$$

with a single learned log-temperature τ. Blocks larger than 1×1 do not commute, so mixed-relation chains are representable, and the operator of a chain is the product of the blocks.

The two dials. k and b are set on validation per graph (Table 1). On `ogbl-biokg`, $k = 12$ and $b = 4$: 27,124,129 parameters counting each complex coefficient as two reals, the convention the board uses for ComplEx (13,562,065 parameter elements). On `ogbl-wikikg2`, $k = 8$ and $b = 64$, i.e. one dense 64×64 operator per relation and direction: 328,842,753 parameters, of which 320M are the table and 8.8M the operators. The reason the dials differ is measured, not chosen (Section 6.3): on the biomedical graph the block-size curve is flat from 4×4 to 16×16 and falls at a dense operator, and width $k = 8$ costs 0.02 whatever the block; on the Wikidata graph the curve rises monotonically to the dense operator, and $k = 8$ with a dense block matches $k = 12$ with 4×4 blocks at 45% of the parameters.

Training shell. The table is stored through its real view as an $(N, 2M)$ tensor and read with a sparse embedding lookup, so a step touches exactly the rows in the batch and autograd emits a row-sparse gradient; reinterpreting the gathered $(B, M, 2)$ rows as complex is free, and every downstream operation is unchanged complex arithmetic. The table is trained with row-wise Adagrad [5, 16]: one accumulator per row, no momentum, untouched rows left bit-identical. The operators and τ use Adam [14] at $5 \cdot 10^{-3}$; a sparse-aware global-norm clip covers both. On `ogbl-wikikg2` at $k = 12$ (a 2.68 GB table) this shell runs at 3.0 ms per step against 64.1 ms for dense Adam on the same batch, with identical validation probes at 1k/2k/3k steps (0.2863/0.3548/0.3706 vs 0.2839/0.3594/0.3719); the dense arm is bound by streaming the table and two moments every step. A self-test asserts the sparse path's scores and gradients equal the dense parent's to 10^{-6}. The storage dtype of the table is a separate choice: bf16 storage (rows upcast to fp32 for all arithmetic) costs nothing measurable on `ogbl-wikikg2` at 200k steps

(0.6890 vs fp32 0.6879) and halves the table; the rows reported here were trained in fp32 and the released teachers are stored in bf16.

Data pipeline. On `ogbl-biokg` entity ids are per type; we flatten them with per-type offsets so each relation's head and tail types map to contiguous ranges. A batch is 2,048 triples of one relation (relations sampled by frequency), scored in one random direction against a shared set of 4,096 negatives drawn uniformly from the target type's range—the Evaluator's own candidate distribution. On `ogbl-wikikg2` there are no types: batches mix relations, each row has its own direction, negatives are uniform over all N, and the training triples live on the GPU so sampling is three integer draws per step. The head-query fraction (0.75 on wikikg2) is a data choice, not architecture: the benchmark's difficulty is almost entirely head prediction (tail queries 0.936, head queries 0.440 for a single model on validation), and 0.75 is the peak of a curve (0.5: 0.6879, 0.75: 0.6968, 0.9: 0.6893, 1.0: 0.2891 at 200k steps). `ogbl-biokg` is direction-symmetric (0.820 / 0.806) and keeps 0.5.

3 Retrieval members from the training graph

The Evaluator scores each query $(h, r, ?)$ over 501 candidates: the true answer and 500 fixed negatives (and symmetrically for heads). For every candidate c we form members from the *holders* of c under r in the training graph, $\mathcal{H}_r(c) = \{h' : (h', r, c) \in \mathcal{T}_{\text{train}}\}$, and the relation of the query entity h to those holders. All members read training edges only; validation and test edges are never used as graph input.

Both boards.

$$\text{analogy}(h, r, c) = \max_{h' \in \mathcal{H}_r(c)} \cos(\text{cnorm}(E_h), \text{cnorm}(E_{h'}))^3, \tag{3}$$

under the model's own embeddings, and its top-3 variant (the mean of the three largest similarities). This is case-based reasoning in the model's own geometry: the embedding compresses an entity's neighbourhood at training time, the graph remembers it exactly at query time. A candidate with no holders gets the minimum score.

`ogbl-biokg`. The Jaccard pair replaces the cosine by raw edge-overlap similarity $|\mathcal{N}(h) \cap \mathcal{N}(h')|/|\mathcal{N}(h) \cup \mathcal{N}(h')|$ over typed training-edge sets; it does not use the model and is computed once. Alone on validation the four members reach 0.645 (analogy), 0.751 (top-3), 0.671 (Jaccard), 0.764 (Jaccard top-3). On the dense typed graph these carry the neighbourhood evidence; two-hop members added nothing (held-out 0.8434 $\to$ 0.8434).

`ogbl-wikikg2`. Uniform negatives on a sparse graph mostly fail on "does c take this relation at all", and the true head of a hub tail ("human") is a rare entity the table knows little about. Six model-free members are shared by all seeds: *holders* $= \log(1 + |\mathcal{H}_r(c)|)$; over the undirected relation-agnostic training graph (15.2M edges, mean degree 12.2) the common-neighbour count *cn*, its Adamic–Adar weighting *cn_aa* (hubs weigh ≈ 0), *linked* (any other training edge between h and c), Adamic–Adar-weighted length-3 paths *cn3_aa*, and *typed* two-hop paths. Alone they are weak (cn_aa 0.096 tail / 0.233 head), together with the per-seed analogy pair they lift a 400k-step model from 0.7001 to 0.7384 held-out. Wikidata hubs with 10^6 holders are capped at 128 holders drawn at random.

Self-augmented members (`ogbl-wikikg2`). Entities whose citizenship, occupation, native language or birthplace is missing from the training graph give the path members nothing to work with. `augment_graph.py` lets the seed-0 model fill them: for each relation, every plausible head (same "instance of" class as the relation's training heads) with no edge under it gets the model's top-1 tail from the relation's candidate set when its confidence clears a per-relation threshold. The threshold and the set of qualifying relations are calibrated on validation labels (the lowest confidence at which the rule's cumulative precision on validation rows of that relation is still ≥ 0.7; relations with fewer than 50 validation rows are skipped): 18 relations pass and 4.18M edges are proposed, e.g. native language 1.02M at precision 0.70, official language at 0.89. No validation or test edge is added; the members are then recomputed on the training graph plus these proposals. Base and augmented members are always offered together so the blend can ignore either per relation: with augmented members alone, sport queries fell ($0.511 \to 0.457$); with both, everything recovers. Effect on the board: row B $0.6820 \to$ B+ 0.6866 test over ten seeds. This is self-training on the training graph; its calibration is a hyperparameter choice on validation, disclosed as such, and its consequence for the validation numbers is handled in Section 5.

Blend. For each member the 501 candidate scores of a query are z-normalised and combined with one weight vector per (relation, direction) group. On `ogbl-biokg` the vector is *selected* from a fixed list of at most nine candidates (uniform means of the top members and softmax weightings by validation MRR) hierarchically—global, relation family, relation—with a group receiving its own weights only above 4,000 validation rows; a learned combiner gains ≤ 0.0007 there because every member already carries a positive weight. On `ogbl-wikikg2` the weights are *learned*: a listwise logistic regression per group over member z-scores (softmax cross-entropy over the 501 candidates, L2 10^{-3}, global $\to$ direction $\to$ relation with warm starts, a group with fewer than 250 fit rows falling back to its direction's weights). It gains +0.026 held-out over selection because wikikg2's members conflict and need negative weights (*holders* is -0.23 on head queries: a candidate that already has many heads is less likely to be the missing one), which selection cannot express. Both procedures are measured cross-fitted—weights fit on a random half of the validation triples, a triple's two directions kept together, scored on the other half—and then refit on the full split and applied once to test.

4 Distillation at training time

Ten checkpoints are a deployment cost, and a uniform score average of ten `ogbl-biokg` models reaches only 0.8428 (one test shot, retired). Instead a fresh model of the same recipe is trained with the added loss

$$\lambda\, T^2\, \mathrm{KL}\big(\mathrm{softmax}(\bar{\ell}/T) \,\big\|\, \mathrm{softmax}(\ell/T)\big) \tag{4}$$

on every batch, where ℓ are the student's logits over [positive | negatives] and $\bar{\ell}$ the mean of the ten frozen teachers' logits on the same candidates [10], $\lambda = 1$; the teachers are the ten row-A seeds and see only the training triples the student sees. The temperature is the finding. On `ogbl-wikikg2`, $T = 1$ transferred nothing (students 0.6669 and 0.6672 test against a plain seed's 0.6676) and $T = 2$ transferred +0.018 (0.6855 ± 0.0008). Brought back to `ogbl-biokg` under a pre-registered +0.003 bar, student-alone validation for seed 0 is 0.8285 / **0.8322** / 0.8314 / 0.8300 at $T = 1/2/3/4$: unimodal with the peak at 2. Our teachers are sharp (the learned scale $e^{\tau} \approx 9$), so at $T = 1$ the soft targets are nearly one-hot over 4,097 candidates and the ordering among near-miss negatives is invisible; $T = 2$ exposes it, $T \geq 3$ flattens it into the cross-entropy's way. Ten `ogbl-biokg` students at $T = 2$ reach 0.8321 ± 0.0003 alone on validation, +0.016 over their teachers.

5 Protocol

Data and metric. `ogbl-biokg`: 93,773 entities of five types, 51 relations, 4,762,678 training, 162,886 validation and 162,870 test triples; filtered MRR over 500 type-matched negatives per direction. `ogbl-wikikg2`: 2,500,604 entities, 535 relations, 16,109,182 training, 429,456 validation and 598,543 test triples; MRR over 500 uniform negatives per direction. Both via `ogb.linkproppred.Evaluator` (`ogb` 1.3.6); our loaders assert that the three splits are pairwise disjoint as loaded.

Pre-registration and test reads. Every configuration's expected test range was written in the project log before its first test read; every comparison between recipes, members, blend rules, temperatures and shells was made on validation. The test split is read once per run: row A by the trainer's final evaluation, the blend rows by the committed shot that refits the weights on full validation and applies them once; test score caches are written without computing any per-member test metric. On `ogbl-biokg`, nine development reads were made outside the rows, each a single shot of a configuration chosen on validation and none re-run: the 25k-step baseline (0.8024); 50k steps with 2×2 blocks, seeds 0 and 1 (0.8084, 0.8080); the final recipe's seeds 0 and 1 trained on a GTX 1080 Ti (0.8134, 0.8123; their RTX 5090 retrains are the first ladder's row A seeds 0 and 1); the uniform ten-model ensemble (0.8428); seed 0 with three analogy features (0.8368); a 48M single model ($k = 16$, 16,384 negatives, per-entity bias) with its features (0.8472); and the same 48M recipe distilled from twelve checkpoints (0.8535). The sparse ladder added none: its forty runs (rows A, B, C at $T = 1$, C at $T = 2$) are the first and only reads of those runs. On `ogbl-wikikg2`, all seed 0 unless noted: two gap diagnostics (200k steps without N3 0.6637, with N3 0.6487); row B's first freeze at a blend guard of 4,000 rows on seeds 0, 1 and 5 (0.6785 / 0.6795 / 0.6791), **re-read** after the guard was lowered to 500 on validation (0.6830 / 0.6848 / 0.6805)—the only runs in this report read twice, and row B is not a submitted entry; $T = 1$ student pilots, seeds 2 and 0 (0.6669 / 0.6672); the ten-seed ensemble with six frozen members (row E, 0.7129) and with augmented and typed members (E+, 0.7164); and a members-only diagnostic (eight shared members under the learned combiner, no model: 0.6723). Each submitted `ogbl-wikikg2` run—the ten students' own final evaluations, the ten C-F blends, the ten leave-one-out ensembles and the full ensemble—was read exactly once.

Validation numbers. For rows whose weights are fit on the full validation split we report the in-sample value and the cross-fitted estimate of the same procedure. On `ogbl-biokg` the estimate is within 0.001 of test on every run. On `ogbl-wikikg2` the learned combiner overstates by ≤ 0.001 in-sample (ensemble 0.7998 vs 0.7992 held-out; C-F 0.7880 vs 0.787 over the seven seeds whose member caches survived; row F 0.7800 vs 0.779), and the self-augmentation thresholds are themselves calibrated on validation, so the held-out figures are the ones filed. The validation–test gap on `ogbl-wikikg2` (0.70 vs 0.67 for a single model) is a relation shift, not a fit: "languages spoken" is 0.4% of validation queries and 21.5% of test, "native language" 0.1% and 7.4%; reweighting validation to the test relation mix reproduces most of the gap (0.6791 vs 0.6678 measured for seed 0).

Statistics. Mean $\pm$ unbiased standard deviation (`torch.std`) over seeds 0–9. Row C's ten students share the same ten teachers, so their deviation reflects student-seed variance only. The ensemble entry's statistics are over ten leave-one-out ensembles of nine seeds each, each read once; they share nine of ten members pairwise, so the deviation reflects membership sensitivity only. The full ten-seed ensemble was read once.

Hardware. The sparse `ogbl-biokg` ladder and all `ogbl-wikikg2` rows were trained on rented RTX 5090 (32 GB) instances (torch 2.14, CUDA 13); the first `ogbl-biokg` ladder on RTX 5090 and GTX 1080 Ti (torch 2.6.0, CUDA 12.4); verification on the 1080 Ti. Checkpoints trained on one machine re-evaluate on the other to within $3 \cdot 10^{-4}$ MRR; the converted sparse-shell seed 0 re-evaluates through the dense code path to its logged test MRR exactly. The biokg analogy

Table 2: `ogbl-biokg`, the submitted (sparse-shell) ladder and the first (dense-Adam) ladder. Ten runs per row, one test read per run. "held-out" is the blend procedure fit on half of validation and measured on the other half; "valid" is the in-sample value. Every row has 27,124,129 parameters.

	test MRR	valid MRR	held-out	hits@1	hits@3	hits@10
sparse shell (submitted)						
A. single model	0.8158 ± 0.0006	0.8164 ± 0.0006	—	0.746	0.866	0.941
B. A + retrieval features	0.8463 ± 0.0004	0.8472 ± 0.0004	0.8465 ± 0.0004	0.788	0.888	0.952
C. distilled ($T = 2$) + retrieval	$\mathbf{0.8528 \pm 0.0002}$	0.8537 ± 0.0002	0.8532 ± 0.0003	0.796	0.893	0.954
C at $T = 1$ (ablation)	0.8509 ± 0.0003					
dense Adam (first ladder, 3 September 2026)						
A. single model	0.8118 ± 0.0013	0.8124 ± 0.0013	—	0.741	0.862	0.939
B. A + retrieval features	0.8425 ± 0.0007	0.8434 ± 0.0007	0.8429 ± 0.0007	0.783	0.884	0.951
C. distilled ($T = 1$) + retrieval	0.8505 ± 0.0004	0.8515 ± 0.0004	0.8512 ± 0.0005	0.794	0.890	0.952

Table 3: Public leaderboard for `ogbl-biokg` as fetched on 2026-09-04 [20], and where the submitted rows sit.

	method	test MRR	params
1	RelEns [28] (ensemble)	0.9618	849M
2	ComplEx^2	0.8583	188M
3	UniBi	0.8550	182M
4	AutoBLM-KGBench [32]	0.8536	192M
	ResonatE C: distilled 27M + retrieval	**0.8528**	**27M**
5	ComplEx-RP [4]	0.8492	188M
	ResonatE B: 27M + retrieval	**0.8463**	**27M**
6	TripleRE [26]	0.8348	470M
7	AutoSF [31]	0.8309	94M
8	PairRE [3]	0.8164	188M
	ResonatE A: 27M single model	**0.8158**	**27M**
9	ComplEx [23]	0.8095	188M
10	DistMult [24]	0.8043	188M
11	RotatE [22]	0.7989	188M
12	TransE [2]	0.7452	188M

features need about 6 GB of RAM and 25–40 minutes per split per seed on one CPU core; the wikikg2 members are GPU-batched and take seconds to minutes per split.

6 Results

6.1 `ogbl-biokg`

Table 2 gives the submitted ladder above the first one. The three clusters of ten do not overlap and each step (+0.031, +0.007) is an order of magnitude larger than the seed noise; the held-out estimate is within 0.0007 of test on every one of the twenty blend runs. The shell is worth +0.0040 on the single model with half the seed spread and +0.0038 with retrieval; with $T = 1$ students the shell's gain is gone at the top of the ladder (0.8509 vs 0.8505, within noise), and the temperature restores it (+0.0019 on top). Table 3 places the rows: C is fifth of twelve and the fourth single model, 0.0008 behind the fourth entry at one seventh of its parameters; every ResonatE row is the smallest model on the board by 3.5×.

Table 4: `ogbl-wikikg2`, ten runs per row, one test read per run. “valid” is in-sample for the blend rows; “held-out” is the cross-fitted estimate of the learned combiner (ensemble and row F: all ten seeds; C-F: the seven seeds whose member caches survived). The single-model rows have 328,842,753 parameters.

	test MRR	valid MRR	held-out
A. single model	0.6676 ± 0.0010	0.7030 ± 0.0009	—
B. A + 8 members, frozen blend	0.6820 ± 0.0014	0.7413	—
B+. + self-augmented members	0.6866 ± 0.0015	0.7467	—
F. A + 10 members + learned combiner	0.7222 ± 0.0010	0.7800	0.779
student alone ($T = 2$)	0.6855 ± 0.0008	0.7190	—
C-F. distilled ($T = 2$) + members + combiner	$\mathbf{0.7320 \pm 0.0010}$	0.7880	0.787
ensemble of 10 seeds + members + combiner	$\mathbf{0.7426 \pm 0.0002}$ (LOO)	0.7994	0.7992
full ten-seed ensemble, one read	0.7430	0.7998	

Table 5: Top of the public leaderboard for `ogbl-wikikg2` as fetched on 2026-09-04 [1] (28 entries), and where the submitted rows sit.

	method	test MRR	params
	ResonatE ×10 + members + combiner	**0.7426**	**3.29B**
1	RelEns [28] (ensemble)	0.7392	2.18B
	ResonatE C-F: distilled 329M + members + combiner	**0.7320**	**329M**
2	StarGraph + TripleRE + Text [17]	0.7305	1.93B
3	InterHT+ [27]	0.7293	156M
4	StarGraph + TripleRE [17]	0.7286	93M
5	InterHT+ (256-dim)	0.7257	148M
6	StarGraph + TripleRE (2022)	0.7201	87M
7	CompoundE3D [7]	0.7006	751M
8	TranS [30]	0.6939	38M
14	ComplEx-RP (50-dim) [4]	0.6392	250M
23	ComplEx (250-dim)	0.4027	1.25B

6.2 `ogbl-wikikg2`

Table 4 gives the ladder. A single 329M model trained in 36 minutes reaches 0.6676; the members and the learned combiner add +0.055; the $T = 2$ student adds +0.010 more; the ten-seed ensemble under the same combiner +0.011 beyond that. Table 5 places the two submitted rows against the public board’s top entries: the ensemble would be first and is also the largest entry on the board (ten tables, 3.29B parameters); the single model C-F would be second, above the best entry that uses entity text and above the best single model, at a mid-table parameter count.

6.3 Why the dials differ

Table 6 is the evidence. On `ogbl-biokg`, width is the binding lever ($k = 8$ costs 0.02 at every block size) and the block size is flat from 4×4 to 16×16 at both widths, then falls at the fully dense operator (-0.013 at $k = 8$). On `ogbl-wikikg2`, capacity in the table is expensive (each unit of k costs millions of rows) and the operator is cheap (535 relations), so larger blocks keep paying all the way to dense, and $k = 8$ with a dense block matches $k = 12$ with 4×4 blocks at 45% of the parameters. One shared setting would cost one of the graphs; the shared setting closest to free is $k = 12$ with 16×16 blocks (biokg 0.7891 vs 0.7893; wikikg2’s best $k = 12$ arm), which we did not run as a submission.

Table 6: Block-size and width curves on validation, seed 0. Left: `ogbl-biokg` at the 12,500-step screen (dense-Adam shell). Right: `ogbl-wikikg2` at 200k steps.

`ogbl-biokg` k, block	params	valid	`ogbl-wikikg2` k, block	params	valid
8, 2×2	12.0M	0.7616	8, 8×8	321M	0.6876
8, 4×4	12.1M	0.7677	8, 16×16	322M	0.6892
8, 16×16	12.2M	0.7677	8, 64×64 (dense)	329M	**0.6945**
8, 64×64 (dense)	12.8M	0.7548	12, 2×2	721M	0.6818
12, 4×4	27.1M	**0.7893**	12, 4×4	721M	0.6879
12, 8×8	27.2M	0.7904	12, 8×8	723M	0.6937
12, 16×16	27.5M	0.7891	12, 16×16	725M	0.6961
16, 4×4	48.2M	0.7928	12, 144×144 (dense)	765M	0.6907
			16, 2×2	1.28B	0.6966

6.4 How the recipe was found

`ogbl-biokg`, first ladder. The starting point was a hand-picked recipe from smaller graphs: $k = 12$, 2×2 blocks, 4,096 negatives, learning rate $5 \cdot 10^{-3}$ cosine, 25,000 steps. Each default was treated as a hypothesis under a tiered protocol written down before each run: a 12,500-step *screen* on validation, promotion only above the current recipe by +0.004 MRR, then a 50,000-step *confirmation*. Sixteen levers went through it; two survived. *Budget*: validation MRR at 6,250 / 12,500 / 25,000 / 50,000 steps is 0.7351 / 0.7829 / 0.8032 / 0.8085, and the gain per doubling shrinks fast enough that 100k projects at $\leq +0.002$, so the budget was doubled once and frozen. *Capacity*: in 2,000-step probes validation MRR grows log-linearly in k with no plateau, but the gain compresses with budget—$k = 16$'s +0.025 at 2k steps is +0.003 at 50k, and even $k = 32$ (192M) misses the bar—whereas $2 \times 2 \rightarrow 4 \times 4$ blocks (118k parameters) keep +0.0055 at 50k. The remaining fourteen levers, mostly the tricks other entries rely on, are in Table 7 (Section 6.5).

`ogbl-wikikg2`. The biokg recipe was run unchanged on the large graph through the new shell (one day, one RTX 5090): a fail-fast ladder of 3k / 20k / 100k / 200k / 800k steps reached TripleRE's level after 67 seconds of training (0.5792 at 20k) and 0.7041 validation at 800k. The dials were swept at 200k (Table 6); the head-query fraction was swept (Section 2); relation-balanced sampling was rejected (up-sampling a relation with 56k triples memorises it: 0.6871 vs 0.6945); N3 lowered both splits equally; the blend guard was lowered from 4,000 to 500 rows after a monotone held-out sweep (0.7392 $\rightarrow$ 0.7411, the test-dominant relations had been denied their own weights); the members were added in the order holders/analogy (0.7001 $\rightarrow$ 0.7224 held-out), two-hop (0.7349), three-hop and a direction level (0.7384), self-augmented (0.7502 with base and augmented offered together), typed paths; the learned combiner replaced selection (0.7499 $\rightarrow$ 0.7791 held-out for the single model); $T = 1$ distillation failed and $T = 2$ succeeded. A compositional entity table (rows built from neighbours through the inverse operators) was piloted on both graphs and rejected: neutral with the table kept, -0.1 to -0.17 without it.

Back-transfer. Two things came back to `ogbl-biokg` under pre-registered bars. The shell: with the row-A recipe unchanged, validation went from 0.8124 ± 0.0013 (dense Adam, ten seeds) to 0.8158 ± 0.0005 (three seeds at table learning rate 0.3; 0.8135 at 0.6), so a ten-seed test row was run with the bar "mean ≥ 0.8143"—met by the mean and by every seed. The temperature: a single $T = 2$ student beat the $T = 1$ student by +0.0037 on validation against a +0.003 bar, and $T = 3$ and $T = 4$ were below $T = 2$. Two things did not: the learned combiner ($\leq +0.0007$ on biokg, whose members never conflict) and two-hop members (no change on a dense typed graph).

Table 7: Levers rejected on `ogbl-biokg` validation (dense-Adam shell). Screen: 12,500 steps on the 4×4 recipe (reference 0.7893); confirm: 50,000 steps (reference 0.8140). "✓" marks a passed screen.

lever	screen	confirm	verdict
relation-prediction aux loss, $\lambda = 0.25/1/4$	0.7886/0.7849/0.7741	—	monotone negative; type-grouped batches already separate relations
N3 regularisation, $w = 10^{-3}/10^{-2}/10^{-1}$	0.7902/0.7912/0.7894	—	real interior peak, half the promotion bar
compositional (path) training, $p = 0.25/0.5$	0.7711/0.7475	—	hops compose exactly already; chains tax the update budget
16,384 negatives	0.7963 ✓	0.8148 (+0.0008)	convergence-speed effect
tied reverse operators (adjoint)	0.7936 ✓	0.8128 (−0.0012)	reverse blocks are not slack
one-cycle schedule, max lr $3 \cdot 10^{-2}$	0.7474†	—	update count is the bottleneck
uniform 8×8 blocks	0.7904	—	flat; 4×4 is the knee
unitary (phase-only) relations	< 90% at every depth (synthetic)		off-circle moduli are load-bearing
real-valued model, equal parameters	0.7923 (2 seeds)	—	"complex" is not a claim
50k budget (adopted)	—	0.8085 vs 0.8032 (25k)	
4×4 blocks (adopted)	0.7893 ✓	0.8140 ✓	survives the budget
sparse shell (adopted)	—	0.8158 vs 0.8124 (3 vs 10 seeds)	then +0.0040 on test, ten seeds

†screened on the 2×2 baseline against its own cosine 0.7829.

6.5 What did not work

Two patterns account for the rejected `ogbl-biokg` levers (Table 7; the 12,500-step reference is the 4×4 recipe's own 0.7893). First, *the architecture pre-empts the trick*: a relation-prediction auxiliary loss [4] is monotonically harmful because type-grouped batches already force the model to separate relations; N3 regularisation [15] has a real but tiny interior optimum because renormalising after every hop already disciplines magnitudes; compositional path training [8] has nothing to repair because hops compose exactly by construction and the chains only spend the update budget. Second, *screen-tier wins vanish at the plateau*: more negatives and tied reverse operators both pass the screen and are flat or negative at 50k—they speed convergence without raising the ceiling. A one-cycle schedule loses to cosine, and phase-only (unitary) relation operators never reach 90% accuracy at any depth in the synthetic chain testbed: the moduli off the unit circle are load-bearing.

Where the errors are. On `ogbl-biokg`'s rarest relations the lookup beats the embedding outright: on side effects with at most 19 training edges the analogy feature alone reaches held-out MRR 0.0175 against the 27M model's 0.0147 (blend 0.0196). Of the remaining errors 36% are drug→side-effect tails, where even a 22-member ensemble scores 0.17—a floor set by the data. Across the ten seeds the truth-versus-runner-up margin has a constant spread of 0.28 and only its mean moves: seeds disagree on which near-tied candidate wins, not on what they know, which is why more seeds were never going to close the gap and why the deployable unit is one table plus a lookup. On `ogbl-wikikg2` 92% of the validation deficit is on head queries, and the members that help most are the ones that answer "who speaks language X" through person → citizenship → country → official language, which is exactly the two-hop path member.

7 One table for everything

The boards are one-hop scores; the architecture was built for chains, and the same table is used, unchanged, for the things below.

Depth. On synthetic worlds with exact ground truth at every depth, models trained only on chains of depth ≤ 4 ($k = 8$, 25k steps, three seeds, 2,000 chains per depth) are read out to depth 48. Three results, in decreasing order of what they show. *One relation applied n times*, on 1,000 entities with seven random permutations as relations: the diagonal model holds 0.987 at depth 8 and crosses 90% between depth 10 and 12 and 50% between 16 and 20; the 4×4-block model 0.963 at depth 8. This is a statement about powers of a single operator, not about composition. *A different relation at every hop*, so that order matters, on the same random permutations: neither the block model nor a dense 64×64 operator learns depth 2 (0.21 and 0.65 accuracy, ≈ 0 beyond), because random permutations share no low-dimensional structure; the block model does not compose arbitrary relations. *A different relation at every hop on relations that form a group*: a turtle on a 16×16 torus with forward, back, turn left and right, strafe left and right, and U-turn, i.e. 1,024 (heading, x, y) states and rigid motions generating $\mathbb{Z}_{16}^2 \rtimes \mathbb{Z}_4$, whose irreducible representations are at most four-dimensional. Half of all depth-2 chains change their answer when reordered, and a commuting diagonal model reaches 0.75 at depth 2. The 4×4-block model trained on depth ≤ 4 reaches 0.949 at depth 8, crosses 90% between depth 10 and 12 and 50% between 20 and 24 (seed s.d. ≤ 0.003 to depth 8); the dense operator reaches further (0.995 at depth 8, 0.885 at depth 20), so the block size is a ceiling rather than the mechanism. The allowed statement is therefore that the model composes non-commuting relations beyond its trained depth when the relations form a group with a low-dimensional representation, and not for arbitrary permutations; whether the relations of a real graph are closer to the former or the latter is not measured. Both mixed-relation experiments were pre-registered after a first external review asked for them. Mapping natural-language questions onto chains of relation operators is outside this report.

Latency, flat in depth, and a compiled index. Because the per-hop renormalisation is a scalar, a chain of relation operators collapses into one block-diagonal operator (2,304 bytes for the 2×2 models) and the readout is a real inner product—maximum inner-product search over a real $N \times 2M$ table. On DRKG [12] (97,238 nodes, 5.87M edges, a 28.1M-parameter model) batched multi-hop queries take 0.08–0.11 ms at every depth 1–6 while exact graph traversal grows with fan-out, up to 45× slower at depth 5–6; compiled scores equal hop-by-hop scores to 10^{-5} with identical top-10 sets over 1,867 queries. An HNSW index [13, 19] over an fp16 copy of the table gives recall@10 0.993 against the exact readout on DRKG (848 bytes per row including links) and, on `ogbl-wikikg2`'s 2,500,604 rows (1.98 GB, 852 bytes per row, built in 673 s on 64 cores), 0.994 at `efSearch` 512 with 2.2 ms per single query on CPU; the exact GPU readout over all 2.5M rows is 2.4–3.3 ms. Storing the table in fp16 leaves the top-10 identical on all 3,334 queries over two biomedical graphs; int8 costs a little (top-1 agreement 0.98), int4 breaks retrieval (0.70). The hub-heavy norm distribution of `ogbl-wikikg2` needs the standard inner-product-to-cosine augmentation (an extra coordinate $\sqrt{R^2 - \|x\|^2}$); a raw inner-product build lost hub rows (recall@10 0.864).

Oracle validation and new edges. On Hetionet [9] and DRKG every top answer is checked against an exact graph oracle and labelled as a citable path or as a novel hypothesis; on the trained metapath libraries the top answer is oracle-verified 94.6–100% of the time across all ten DRKG paths and 99–100% on all eight Hetionet paths. On `ogbl-wikikg2` the same readout with the same flags: for Douglas Adams, "notable work" (two training edges) returns *Last Chance to See*, *The Restaurant at the End of the Universe*, *The Salmon of Doubt*, *Mostly Harmless* and *The Hitchhiker's Guide* in the top ten, nine of them absent from every split; "educated at" returns Cambridge and St John's (training edges) and then Harvard, Trinity, York and Oxford as novel. The self-augmentation of Section 3 is the same mechanism run at scale with a validation-calibrated threshold: 4.18M proposed edges whose per-relation precision on held-out validation rows is ≥ 0.7, and which raised every downstream row. Fine-tuning 496 held-out *treats* edges into a trained DRKG model took 3.7 GPU-minutes, absorbed all of them to rank 1 (MRR 0.147 $\rightarrow$ 1.000) with a worst metapath regression of -0.008.

Training cost and memory. A training step touches the rows in its batch whatever N is, and the optimiser keeps one float per row. `ogbl-biokg`: 50,000 steps in 231 s on one RTX 5090 (18 minutes on a GTX 1080 Ti with the dense shell); a student with ten teachers in 506 s. `ogbl-wikikg2`: the 329M model trains 800,000 steps at 2.7 ms per step, 36.5 minutes, from a 1.19 GB fp32 table (0.6 GB in bf16); a student with ten bf16 teachers resident, 400,000 steps at 7.6 ms, 50 minutes; the $k = 12$ table (2.68 GB) trained at 3.0 ms per step in 15.7 GB of GPU memory, against 64.1 ms for dense Adam. Row-wise Adagrad costs nothing measurable on the biomedical graphs either (Hetionet hits@10 0.782 vs 0.782 over three paired seeds; DRKG within one standard error). From the measured per-row costs, 100M entities at $k = 12$ would need a 115 GB fp32 training table, 0.4 GB of optimiser state, an 85 GB fp16 HNSW serving index (or about 5 GB of RAM plus NVMe in a DiskANN layout [21]) and roughly 7 GPU-hours for 20 epochs of 500M edges on one 5090-class card; that remains a projection, and none of it has been run.

8 Limitations and disclosures

The blend weights of the retrieval rows are selected or fit on the full validation split; we report the in-sample number and the cross-fitted estimate, which is within 0.001 of test on `ogbl-biokg` and within 0.001 of in-sample on `ogbl-wikikg2`, where the test split's relation mix differs from validation's and no validation number predicts test. The self-augmentation thresholds on `ogbl-wikikg2` are calibrated on validation labels; no validation or test edge is used as input, but the augmented rows' in-sample validation numbers inherit that calibration. Three of the ten `ogbl-wikikg2` students (seeds 0, 2, 3) were deleted by the campaign script after their reads and cannot be re-verified from a checkpoint; their receipts and logs are released. Row C's students share one teacher pool on both boards; the leave-one-out ensembles share nine members pairwise. The wikikg2 ensemble entry is ten 329M tables, the largest entry on that board. At inference a retrieval row needs the holder lists of a query's candidates—cheap, but a graph lookup the pure embedding does not need. Every development test read is listed in Section 5, including the three runs of a non-submitted row that were read twice. Training is seeded but CUDA kernels are not bit-deterministic, so a retrained checkpoint reproduces its MRR to seed-level noise rather than bit-for-bit; the blend stage is exact. No external data was used on either board.

9 Reproducibility

The repository `https://github.com/jinxmcg/resonate` holds the shared model code at its root (`resonate.py`, the sparse shell `resonate_wiki.py`, `rowadagrad.py`) and one folder per board. `biokg/` contains both ladders' trainers, the member and blend scripts, the receipts of the submitted ladder (`results/sparse`) and of the first (`results/dense`), a `summarize.py` that recomputes every biokg statistic in this report from either, and a `verify.py` that asserts split disjointness and re-evaluates the released checkpoints of either ladder against their logs. `wikikg2/` contains the trainer, the member scripts, the learned combiner, the campaign scripts as run across three rented machines with the machine-specific paths removed, the receipts of every row and the cross-fitted validation logs, and `summarize_wiki.py`. The releases carry the twenty submitted `ogbl-biokg` checkpoints (row-A models re-saved in the dense format by an exact reinterpretation of the real-view table, and the $T = 2$ students), the first ladder's twenty, and `ogbl-wikikg2`'s ten teachers and seven surviving students with bf16 tables.

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